LAPORAN PRAKTIKUM

POSTTEST 2

ALGORITMA PEMROGRAMAN LANJUT

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PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

SAMARINDA

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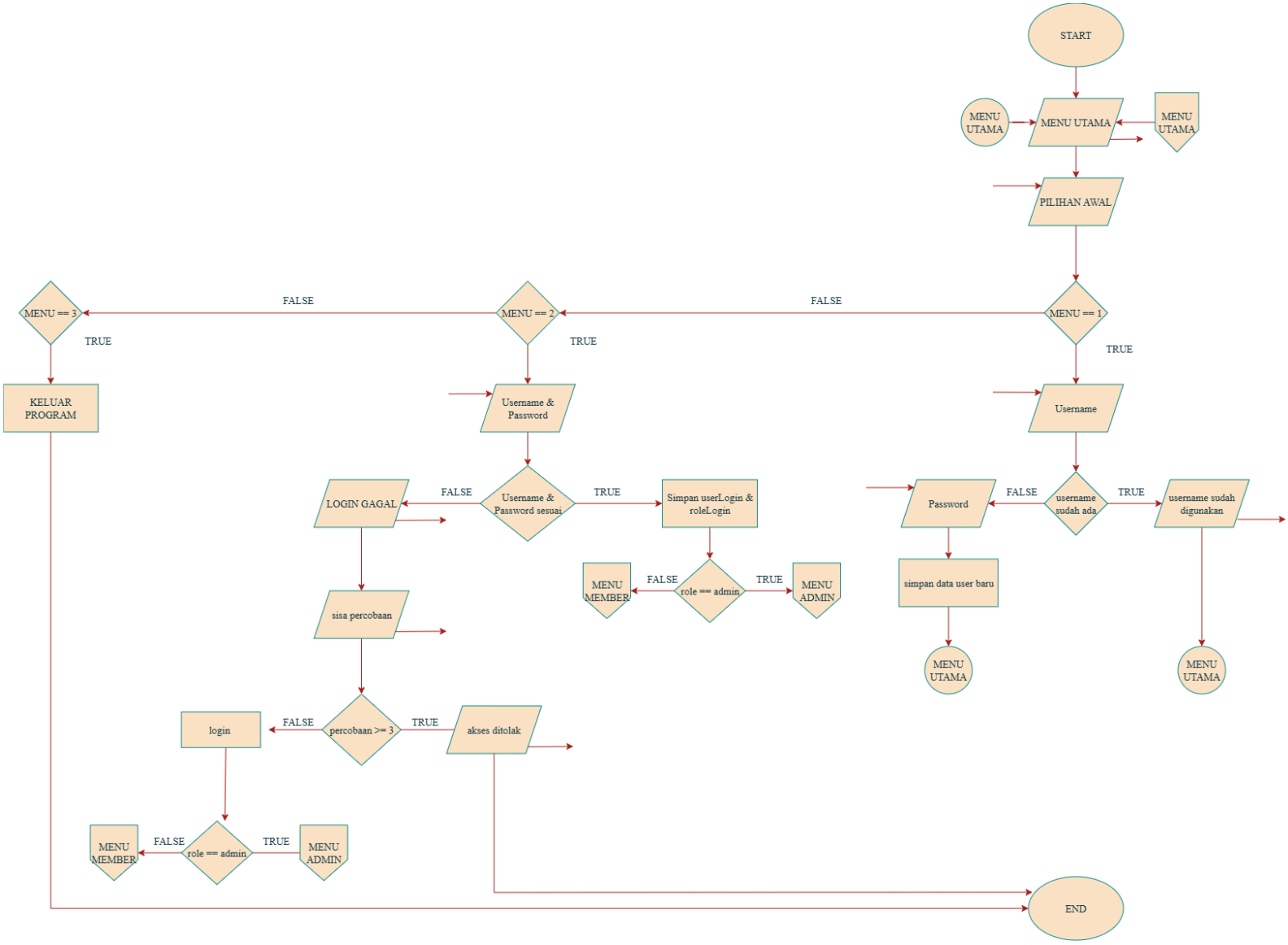
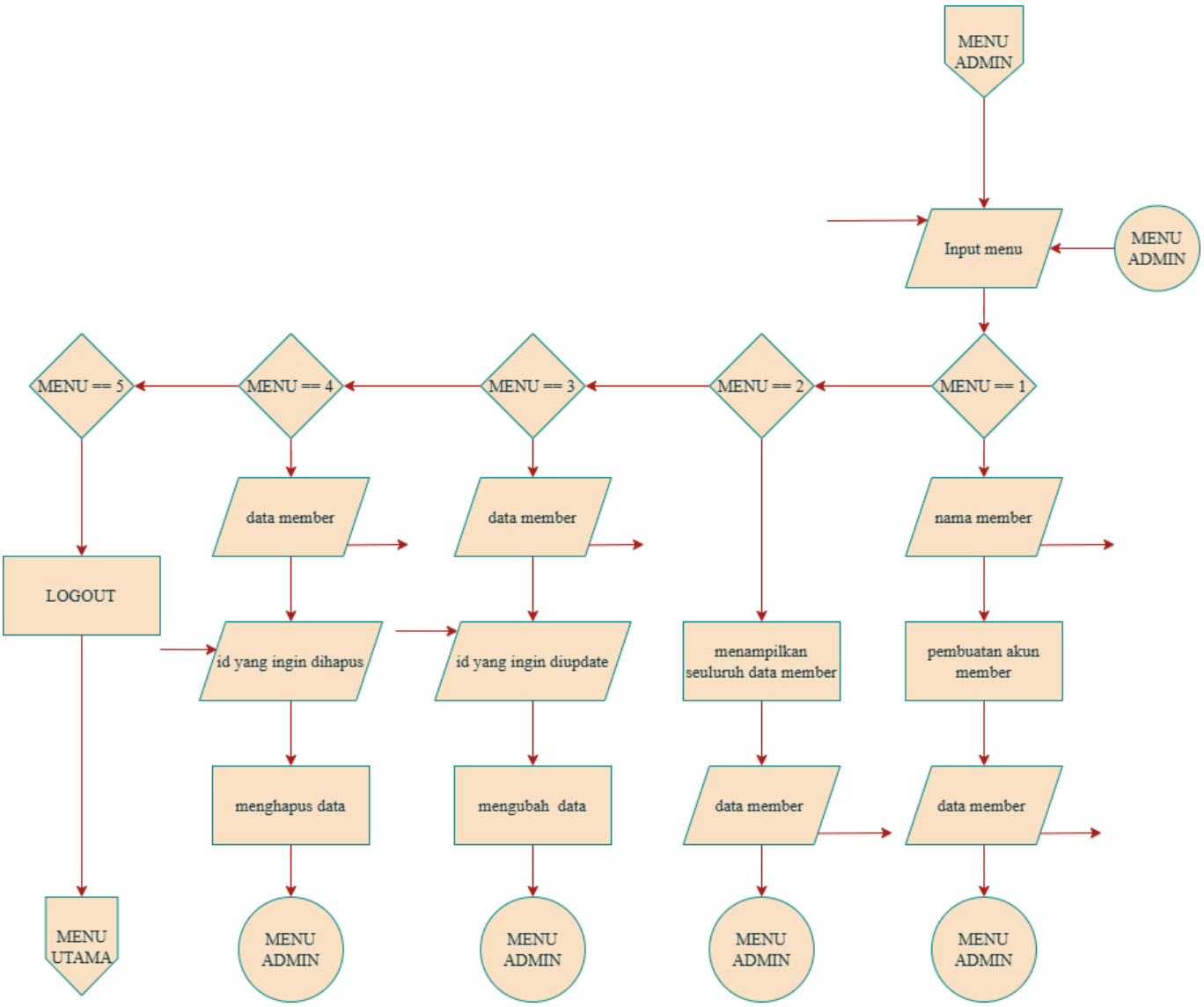
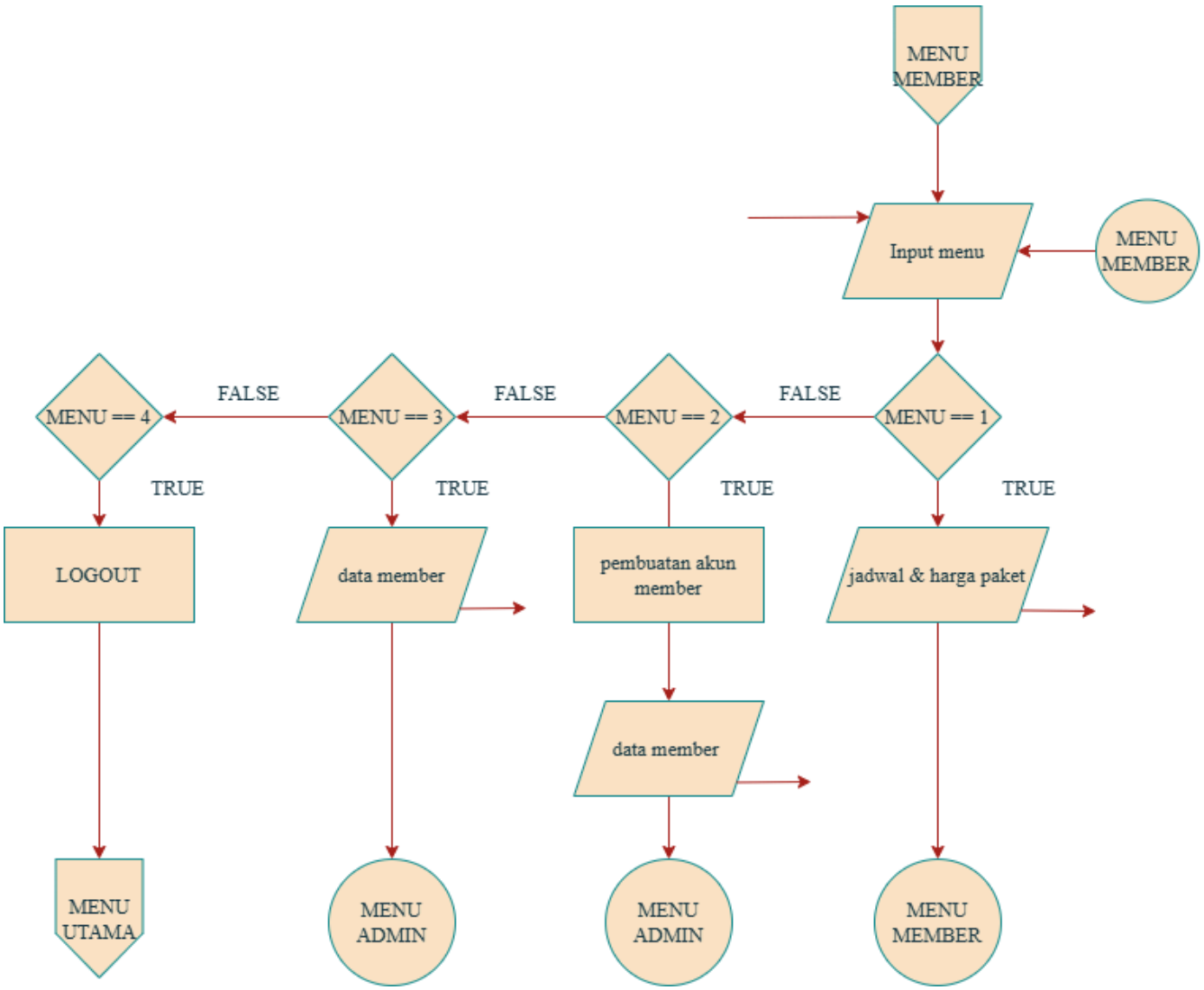
1. Flowchart


START
MENU UTAMA
MENU UTAMA
MENU UTAMA
PILIHAN AWAL
MENU == 1
FALSE
TRUE
MENU == 2
FALSE
TRUE
MENU == 3
FALSE
TRUE
KELUAR PROGRAM
Username
Password
FALSE
TRUE
username sudah ada
username sudah digunakan
simpan data user baru
MENU UTAMA
MENU UTAMA
Username & Password
Username & Password sesuai
FALSE
TRUE
LOGIN GAGAL
Simpan userLogin & roleLogin
role == admin
FALSE
TRUE
MENU MEMBER
MENU ADMIN
sisa percobaan
percobaan >= 3
FALSE
TRUE
login
akses ditolak
role == admin
FALSE
TRUE
MENU MEMBER
MENU ADMIN
END


MENU ADMIN
Input menu
MENU ADMIN
MENU == 5
MENU == 4
MENU == 3
MENU == 2
MENU == 1
data member
data member
nama member
LOGOUT
id yang ingin dihapus
id yang ingin diupdate
menampilkan seuluruh data member
pembuatan akun member
menghapus data
mengubah data
data member
data member
MENU UTAMA
MENU ADMIN
MENU ADMIN
MENU ADMIN
MENU ADMIN



1. **Flowchart Menu Utama**

- START → Program dimulai
- Tampilkan MENU UTAMA dengan 3 pilihan:
    1. Registrasi Akun Baru
    2. Login
    3. Keluar
- PILIHAN AWAL:
    1. Jika MENU == 1 (REGISTRASI):
        - Input Username
        - Cek apakah username sudah ada?

        - TRUE → Tampil "username sudah digunakan" → Kembali ke MENU UTAMA
        - FALSE → Input Password → Simpan data user baru → Kembali ke MENU UTAMA
    2. B. Jika MENU == 2 (LOGIN):
        - Input Username & Password
        - Cek kredensial (username & password sesuai?)
        - FALSE → LOGIN GAGAL → Hitung sisa percobaan
        - Jika percobaan >= 3 → AKSES DITOLAK → Program END
        - Jika percobaan < 3 → Kembali input login
        - TRUE → Simpan userLogin & roleLogin
        - Cek role == admin?
        - TRUE → Masuk ke MENU ADMIN
        - FALSE → Masuk ke MENU MEMBER
    3. Jika MENU == 3 (KELUAR):
        - KELUAR PROGRAM → END

2. Flowchart Menu Admin

- MENU == 1 → Buat Akun Member Baru
    1. Input nama member
    2. Proses pembuatan akun member
    3. Simpan data member → Kembali ke MENU ADMIN
- MENU == 2 → Tampilkan Seluruh Data Member
    1. Menampilkan semua data dalam array
    2. Tampil data member → Kembali ke MENU ADMIN
- MENU == 3 → Update Data Member
    1. Input id yang ingin diupdate
    2. Tampil data member yang akan diubah
    3. Proses mengubah data (paket, hari, jam, durasi)
    4. Simpan perubahan → Kembali ke MENU ADMIN
- MENU == 4 → Hapus Data Member
    1. Input id yang ingin dihapus
    2. Tampil data member yang akan dihapus
    3. Proses menghapus data (shift array)

4. Kembali ke MENU ADMIN

- MENU == 5 → LOGOUT → Kembali ke MENU UTAMA

3. Flowchart Menu Member

- MENU == 1 → Tampilkan jadwal & harga paket → Kembali ke MENU MEMBER
- MENU == 2 → Pembuatan akun member
    1. Input data member (nama, paket, hari, jam, durasi)
    2. Simpan data member → Kembali ke MENU MEMBER
- MENU == 3 → Tampilkan data member (data diri sendiri) → Kembali ke MENU MEMBER
- MENU == 4 → LOGOUT → Kembali ke MENU UTAMA

2. Deskripsi Singkat Program

Program ini dibuat dengan tujuan untuk memudahkan pengelolaan data member kelas padel secara terstruktur dan efisien. Program ini memungkinkan admin untuk mengelola data member (create, read, update, delete) serta member untuk mendaftar dan melihat data diri sendiri, dengan validasi paket sesi padel (SUN/MOON/STAR), pemilihan hari sesuai jadwal, dan perhitungan biaya otomatis berdasarkan durasi member.

3. Source Code

A. MENU UTAMA

```
int main() {
    User users[MAX_USER] = {
        {"risma", "102", "member"},
        {"admin", "milopadel", "admin"}
    };
    int jumlahuser = 2;

    member members[MAX_MEMBER];
    int jumlahmember = 0;

    string inputnama, inputpw;
    string userLogin, roleLogin;

    while (true) {
        cout << "!=====================================!" << endl;
        cout << "        WELCOME TO MILO PADEL        " << endl;
```

```
    cout << "!=====================================!" << endl;

    cout << "\n--- MENU UTAMA---" << endl;
    cout << "1. Registrasi Akun Baru" << endl;
    cout << "2. Login" << endl;
    cout << "3. Keluar" << endl;
    cout << "Pilihan: ";
    int pilihanawal;
    cin >> pilihanawal;
    cin.ignore();

    if (pilihanawal == 1) {
        cout << "\n--- REGISTRASI AKUN ---" << endl;
        cout << "Username: ";
        getline(cin, inputnama);

        bool usernameada = false;
        for (int i = 0; i < jumlahuser; i++) {
            if (inputnama == users[i].username) {
                usernameada = true;
                break;
            }
        }

        if (usernameada) {
            cout << "[!] Username \"" << inputnama << "\" SUDAH DIGUNAKAN!" << endl;
            continue;
        }

        cout << "Password: ";
        getline(cin, inputpw);
        users[jumlahuser].username = inputnama;
        users[jumlahuser].password = inputpw;
        users[jumlahuser].role = "user";
        jumlahuser++;

        cout << "\n[+] REGIS BERHASIL! YUK LOGIN" << endl;
        cout << endl;
    }

    else if (pilihanawal == 2) {
        cout << "\n--- LOGIN USER ---" << endl;

        bool loginSukses = false;

        for (int percobaan = 0; percobaan < 3; percobaan++) {
            cout << "Username: ";
            getline(cin, inputnama);
            cout << "Password: ";
            getline(cin, inputpw);
```

```
        bool usnbenar = false;
        bool pwbenar = false;

        for (int i = 0; i < jumlahuser; i++) {
            if (inputnama == users[i].username) {
                usnbenar = true;
                if (inputpw == users[i].password) {
                    pwbenar = true;
                    userLogin = inputnama;
                    roleLogin = users[i].role;
                    break;
                }
            }
        }

        if (usnbenar && pwbenar) {
            loginSukses = true;
            break;
        }

        cout << "\n[!] LOGIN GAGAL" << endl;
        if (!usnbenar) {
            cout << "    -> USERNAME GA ADA NIH!" << endl;
        } else {
            cout << "    -> PASSWORD SALAH!" << endl;
        }

        if (percobaan == 2) {
            cout << "\n[!] AKSES DITOLAK! GAGAL LOGIN 3x." << endl;
            return 0;
        }

        cout << "    -> Sisa percobaan: " << (2 - percobaan) << endl << endl;
    }

    if (!loginSukses) {
        return 0;
    }

    if (roleLogin == "admin") {
        menuAdmin(members, jumlahmember, userLogin);
    } else {
        menuMember(members, jumlahmember, userLogin);
    }

} else if (pilihanawal == 3) {
    cout << "\nTHANK YOU! see you milovers." << endl;
    break;
} else {
    cout << "[!] PILIHAN TIDAK VALID!" << endl;
}
```

```
    }

    return 0;
}
```

B. MENU ADMIN

```
void menuAdmin(member members[], int &jumlahmember, string userLogin) {
    int pilihan;
    bool menuberjalan = true;

    do {
        cout << endl;
        cout << "=============================================" << endl;
        cout << "          login sebagai: " << userLogin << " (admin)" << endl;
        cout << "          mau ngapain nih???                " << endl;
        cout << "=============================================" << endl;

        cout << "1. Buat Akun Member Baru      " << endl;
        cout << "2. Tampilkan Semua Data       " << endl;
        cout << "3. Update Paket Member        " << endl;
        cout << "4. Hapus Member               " << endl;
        cout << "5. Logout                     " << endl;
        cout << "=========================================" << endl;
        cout << "Pilih menu (1-5): ";
        cin >> pilihan;
        cin.ignore();

        if (pilihan < 1 || pilihan > 5) {
            cout << "[!] PILIHAN TIDAK VALID! hanya 1-5." << endl;
            continue;
        }

        switch (pilihan) {
            case 1:
                buatmember(members, jumlahmember, "admin_managed");
                break;
            case 2:
                bacasemuadata(members, jumlahmember);
                break;
            case 3:
                updatedatamember(members, jumlahmember);
                break;
            case 4:
                deletedatamember(members, jumlahmember);
                break;
            case 5:
                cout << "\n[+] BERHASIL LOGOUT! kembali ke menu utama." << endl;
                menuberjalan = false;
```

```
            break;
        }

    } while (menuberjalan);
}
```

C. MENU MEMBER

```
void menuMember(member members[], int &jumlahmember, string userLogin) {
    int pilihan;
    bool menuberjalan = true;

    do {
        cout << endl;
        cout << "=============================================" << endl;
        cout << "          login sebagai: " << userLogin << " (member)" << endl;
        cout << "          mau ngapain nih???                " << endl;
        cout << "=============================================" << endl;

        cout << "1. Lihat Jadwal & Harga Paket   " << endl;
        cout << "2. Buat Akun Member             " << endl;
        cout << "3. Lihat Data Saya              " << endl;
        cout << "4. Logout                       " << endl;
        cout << "=============================================" << endl;
        cout << "Pilih menu (1-4): ";
        cin >> pilihan;
        cin.ignore();

        if (pilihan < 1 || pilihan > 4) {
            cout << "[!] PILIHAN TIDAK VALID! hanya 1-4." << endl;
            continue;
        }

        switch (pilihan) {
            case 1:
                tampilkaninfopaket();
                break;
            case 2:
                buatmember(members, jumlahmember, userLogin);
                break;
            case 3:
                bacadatasaya(members, jumlahmember, userLogin);
                break;
            case 4:
                cout << "\n[+] BERHASIL LOGOUT! kembali ke menu utama." << endl;
                menuberjalan = false;
```

```
            break;
        }

    } while (menuberjalan);
}
```

4. Hasil Output

```
!=================================!
        WELCOME TO MILO PADEL
!=================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 1

--- REGISTRASI AKUN ---
Username: risma
[!] Username "risma" SUDAH DIGUNAKAN!
!=================================!
        WELCOME TO MILO PADEL
!=================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan:
```

Gambar 4.1 Hasil Output

```
!=====================================!
        WELCOME TO MILO PADEL
!=====================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 1

--- REGISTRASI AKUN ---
Username: dapa
Password: 111

[+] REGIS BERHASIL! YUK LOGIN

!=====================================!
        WELCOME TO MILO PADEL
!=====================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan:
```

Gambar 4.2 Hasil Output

```
!===================================!
        WELCOME TO MILO PADEL
!===================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 2

--- LOGIN USER ---
Username: risma
Password: 102

============================================
          login sebagai: risma (member)
          mau ngapain nih???
============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
============================================
Pilih menu (1-4):
```

Gambar 4.3 Hasil Output

```
--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 2

--- LOGIN USER ---
Username: dapa
Password: 123

[!] LOGIN GAGAL
    -> PASSWORD SALAH!
    -> Sisa percobaan: 2

Username: 124
Password: dapa

[!] LOGIN GAGAL
    -> USERNAME GA ADA NIH!
    -> Sisa percobaan: 1

Username: dapa
Password: 124

[!] LOGIN GAGAL
    -> PASSWORD SALAH!

[!] AKSES DITOLAK! GAGAL LOGIN 3x.
PS C:\Users\lenov\Downloads\praktikum_apl\post-tes
```

Gambar 4.4 Hasil Output

```
!=====================================!
        WELCOME TO MILO PADEL
!=====================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 2

--- LOGIN USER ---
Username: risma
Password: 102

=============================================
          login sebagai: risma (member)
          mau ngapain nih???
=============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=============================================
Pilih menu (1-4): 1

=====================================================
               PAKET SESI DI MILO PADEL
=====================================================
1. SUN   : weekday, jam 06.00-16.00 | Rp 450.000/sesi
2. MOON  : weekday, jam 16.00-23.00 | Rp 500.000/sesi
3. STAR  : weekend, jam 16.00-23.00 | Rp 650.000/sesi
=====================================================
```

Gambar 4.5 Hasil Output

```
[FYI] 1 bulan = 4 pertemuan (1x seminggu)
=====================================================

=============================================
          login sebagai: risma (member)
          mau ngapain nih???
=============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=============================================
Pilih menu (1-4): 2

--- BUAT AKUN MEMBER ---
Nama: risma

=====================================================
                PAKET SESI DI MILO PADEL
=====================================================
1. SUN   : weekday, jam 06.00-16.00 | Rp 450.000/sesi
2. MOON  : weekday, jam 16.00-23.00 | Rp 500.000/sesi
3. STAR  : weekend, jam 16.00-23.00 | Rp 650.000/sesi
=====================================================
[FYI] 1 bulan = 4 pertemuan (1x seminggu)
=====================================================
Paket (SUN/MOON/STAR): sun
[!] MAAF, PAKET INI GA ADA!
```

Gambar 4.6 Hasil Output

```
=============================================
          login sebagai: risma (member)
          mau ngapain nih???
=============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=============================================
Pilih menu (1-4): 2

--- BUAT AKUN MEMBER ---
Nama: risma

=====================================================
                PAKET SESI DI MILO PADEL
=====================================================
1. SUN   : weekday, jam 06.00-16.00 | Rp 450.000/sesi
2. MOON  : weekday, jam 16.00-23.00 | Rp 500.000/sesi
3. STAR  : weekend, jam 16.00-23.00 | Rp 650.000/sesi
=====================================================
[FYI] 1 bulan = 4 pertemuan (1x seminggu)
=====================================================
Paket (SUN/MOON/STAR): SUN
MAU PADEL DI HARI APA (Senin-Jumat): sabtu
[!] HARI TIDAK VALID! Paket weekday hanya bisa: Senin-Jumat
MAU PADEL DI HARI APA (Senin-Jumat): kamis
```

Gambar 4.7 Hasil Output

```
--- PILIH JAM SESI (max 1 pilihan) ---
1. 06.00 - 07.00
2. 07.00 - 08.00
3. 08.00 - 09.00
4. 09.00 - 10.00
5. 10.00 - 11.00
6. 11.00 - 12.00
7. 12.00 - 13.00
8. 13.00 - 14.00
9. 14.00 - 15.00
10. 15.00 - 16.00
JAM BERAPA (1-10): 11

MAU JOIN BERAPA BULAN: 2

[+] YEY AKUN MEMBER UDAH ADA!
      Paket           : SUN
      Hari            : Kamis
      Jam sesi        : -
      Durasi          : 2 bulan
      Pertemuan/Bulan : 4x
      Total Pertemuan : 8x
      Harga /Sesi     : Rp 450.000
      TOTAL BAYAR   : Rp 3600.000

==========================================
          login sebagai: risma (member)
          mau ngapain nih???
==========================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
==========================================
Pilih menu (1-4):
```

Gambar 4.8 Hasil Output

```
Pilih menu (1-4): 3

=====================================
              DATA MEMBER ANDA
========================================
ID               : 1
Nama             : risma
Paket            : SUN
Hari             : Kamis
Jam sesi         : -
Durasi           : 2 bulan
Pertemuan/Bulan  : 4x
Total Pertemuan  : 8x
Harga /Sesi      : Rp 450.000
 TOTAL BAYAR    : Rp 3600.000
========================================

=============================================
          login sebagai: risma (member)
          mau ngapain nih???
=============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=============================================
Pilih menu (1-4): 4

[+] BERHASIL LOGOUT! kembali ke menu utama.
!===================================!
        WELCOME TO MILO PADEL
!===================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
```

Gambar 4.9 Hasil Output

```
=============================================
          login sebagai: risma (member)
          mau ngapain nih???
=============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=============================================
Pilih menu (1-4): 4

[+] BERHASIL LOGOUT! kembali ke menu utama.
!===================================!
        WELCOME TO MILO PADEL
!===================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 3

THANK YOU! see you milovers.
PS C:\Users\lenov\Downloads\praktikum_apl\post-t
```

Gambar 4.10 Hasil Output

```
!=====================================!
        WELCOME TO MILO PADEL
!=====================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 2

--- LOGIN USER ---
Username: admin
Password: milopadel

=============================================
          login sebagai: admin (admin)
          mau ngapain nih???
=============================================
1. Buat Akun Member Baru
2. Tampilkan Semua Data
3. Update Paket Member
4. Hapus Member
5. Logout
========================================
Pilih menu (1-5):
```

Gambar 4.11 Hasil Output

5. Langkah-langkah GIT

5.1 GIT Add

Git Add adalah perintah yang digunakan untuk menambahkan semua file yang ada di folder tersebut ke staging area. Titik (.) berarti semua file yang berubah atau baru dibuat akan disiapkan untuk disimpan.

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\lenov\Downloads\praktikum_apl> git add .
```

Gambar 4.12 Git Add

5.2 GIT Commit

Git Commit adalah Commit menyimpan perubahan dari staging area ke local repository -m "finish pt 2" adalah pesan commit yang menjelaskan perubahan ini untuk "finish pt 2" (pertemuan 2). Ada 6 files changed → Ada 6 file yang

diubah/ditambahkan 900 insertions(+) → Sekitar 900 baris kode baru ditambahkan 1 deletion(-) → 1 baris kode dihapus create mode 100644 → File-file baru dibuat (termasuk file .cpp dan .exe)

```
PS C:\Users\lenov\Downloads\praktikum_apl> git commit -m "finish pt 2"
[main 236a7a5] finish pt 2
 6 files changed, 900 insertions(+), 1 deletion(-)
 create mode 100644 kelas/pertemuan-1/pertemuan-2.cpp
 create mode 100644 kelas/pertemuan-1/pertemuan-2.exe
 create mode 100644 post-test/post-test-apl-2/2509106102-RISMAYANTY-PT-2.cpp
 create mode 100644 post-test/post-test-apl-2/2509106102-RISMAYANTY-PT-2.exe
 create mode 100644 post-test/post-test-apl-2/tempCodeRunnerFile.cpp
```

Gambar 4.13 Git Commit

5.3 GIT Push

Upload commit dari komputer lokal ke GitHub (remote repository) origin = nama remote repository main = branch yang di-push -u = set upstream (menghubungkan branch lokal dengan remote), Artinya ada 19 objects → Total objek Git yang diproses Compressing objects → File dikompres sebelum dikirim (11 file) Writing objects → Upload ke GitHub (13 objek, ukuran 956.79 KiB) remote: This repository moved → Repository pindah ke URL baru bc3ead6..236a7a5 → Commit hash sebelumnya → commit hash baru main -> main → Branch lokal main berhasil di-push ke remote main branch 'main' set up to track 'origin/main' → Branch lokal sekarang terhubung dengan remote

```
PS C:\Users\lenov\Downloads\praktikum_apl> git push -u origin main
Enumerating objects: 19, done.
Counting objects: 100% (19/19), done.
Delta compression using up to 16 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (13/13), 956.79 KiB | 4.16 MiB/s, done.
Total 13 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 2 local objects.
remote: This repository moved. Please use the new location:
remote:   https://github.com/rismayanty/praktikum-apl.git
To https://github.com/rismayanty/praktikum.apl.git
   bc3ead6..236a7a5  main -> main
branch 'main' set up to track 'origin/main'.
```

Gambar 4.14 Git Push